

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 2 Mathematics

TRIAL HIGHER SCHOOL CERTIFICATE

August 2012

TIME ALLOWED: 180 minutes

READING TIME: 5 minutes

General Instructions:

- Write your name and class at the top of this page, and on your answer booklet.
- Hand in all of your answers and this question sheet.
- Use only blue or black pen
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- **START ALL QUESTIONS ON A NEW PAGE**
- Approved calculators may be used.
- A table of *Standard Integrals* is attached. You may detach this page now.

Section I Pages 1 to 5

10 marks

- Colour in the circle on your Section I answer sheet corresponding to the correct answer
- There is only one correct answer for all questions in this section
- Allow about 15 minutes for this section

Section II Pages 6 to 16

90 marks

- Allow about 2 hours 45 minutes for this section

SECTION I

1 If $z = 1 + \sqrt{3}i$, then $z^4 =$

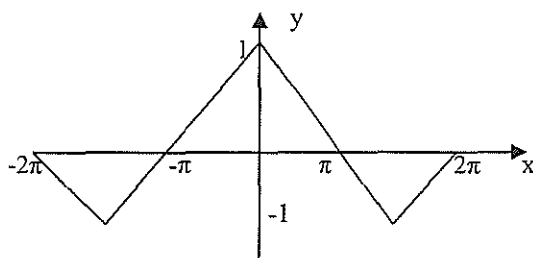
- A** $8 + 8\sqrt{3}i$
- B** $8 - 8\sqrt{3}i$
- C** $-8 + 8\sqrt{3}i$
- D** $-8 - 8\sqrt{3}i$

2 $\int \sin^3 x dx =$

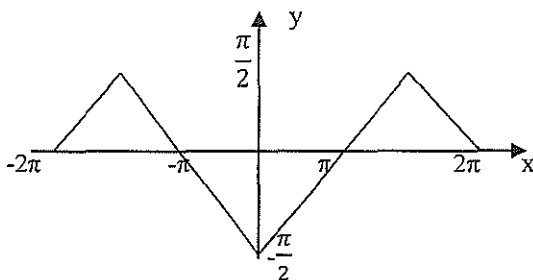
- A** $\frac{1}{4}\sin^4 x + k$
- B** $-\cos x + \frac{1}{3}\cos^3 x + k$
- C** $-\cos x - \frac{1}{3}\cos^3 x + k$
- D** $\cos x - \frac{1}{3}\cos^3 x + k$

3 Which of the curves below represents the curve $y = \sin^{-1}(\cos x)$ for $-2\pi \leq x \leq 2\pi$

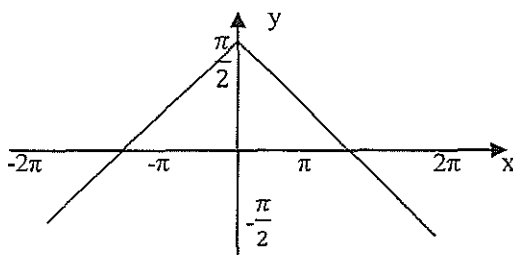
A



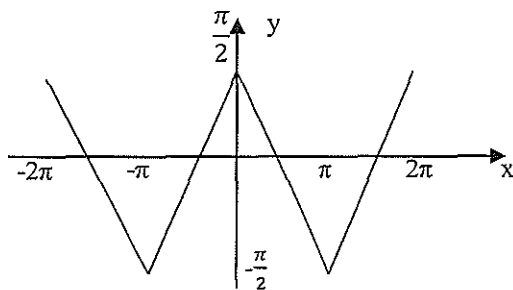
B



C



D



4

Given that $\tan x + \cot x = \frac{1}{\sin x \cos x}$ then a Primitive of $\frac{1}{\sin x \cos x}$ is

A $\frac{1}{\cos^2 x} \log \sin x$

B $\log \sin x \cos x$

C $\log |\tan x|$

D $\log \cot x$

5

A quadratic expression with zeros of $4 + i$ and $4 - i$ is:

A $x^2 - 8x + 17$

B $x^2 + 8x + 17$

C $x^2 - 8x - 17$

D $x^2 + 8x - 17$

6

The derivative of the curve

$$x^3 + 9x^2 - y^2 + 27x - 4y + 23 = 0 \quad \text{is:}$$

A $\frac{dy}{dx} = \frac{x^2+6x+9}{2y}$

B $\frac{dy}{dx} = \frac{x^2+6x+9}{-2y}$

C $\frac{dy}{dx} = \frac{3x^2+18x+27}{-2y-4}$

D $\frac{dy}{dx} = \frac{3x^2+18x+27}{2y+4}$

7

For the curve given by:

$$f(x) = x^2, \quad 0 \leq x \leq 1$$

$$f(x) = 2x - 1, \quad x > 1$$

Which of the following statements is correct?

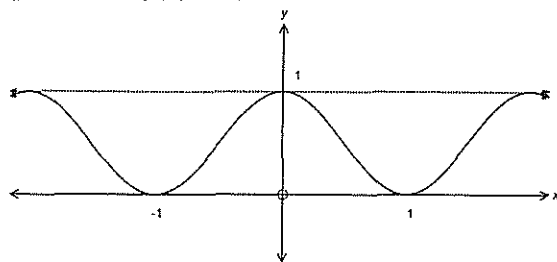
A $f(x)$ has a discontinuity at $x = 1$

B $f(x)$ is differentiable at $x = 1$

C $f(x)$ has an asymptote at $x = 1$

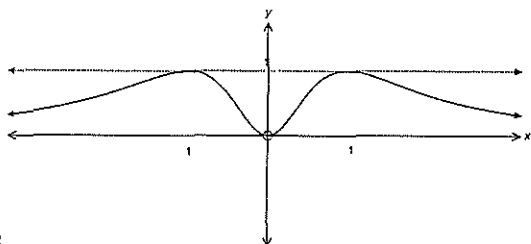
D $f(x)$ has a stationary point at $x = 1$

8 The graph of $y = f(x)$ is given below.

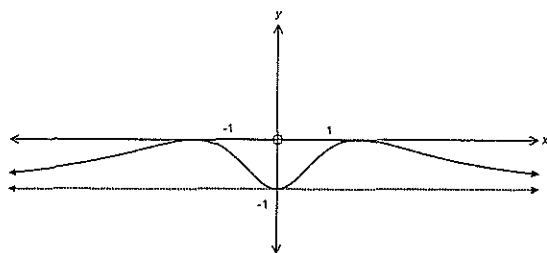


Which of the following represents $y = \frac{1}{f(x)}$?

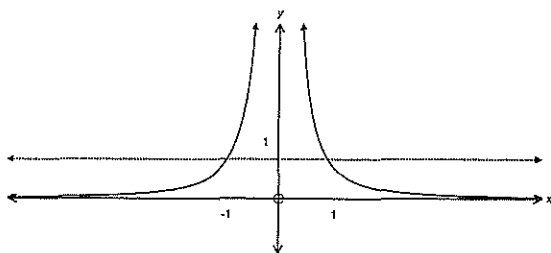
A



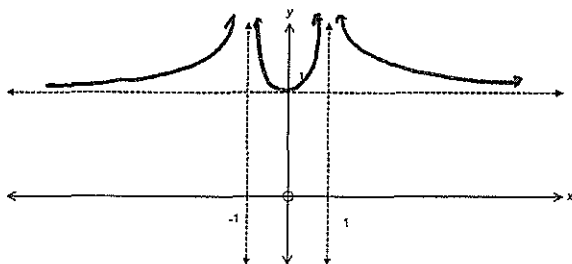
B



C



D



9 It is known that $x = 2 - 3i$ is a solution to $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$

Another solution is $x =$

A $1 - 2i$

B $-1 - 2i$

C $-2 - i$

D $-2 + i$

10 A particle moves in a straight line so that its velocity at any particular time is given by $v = k(a - x)$, where x is its displacement from a given point O.

The particle is initially at O.

Which of the following gives an expression for x :

A $x = a(1 - e^{kt})$

B $x = a(1 + e^{kt})$

C $x = a(1 - e^{-kt})$

D $x = a(1 + e^{-kt})$

SECTION II

QUESTION 11:

Marks

2 (a) Find $\int \frac{dx}{x^2 - 6x + 13}$

4 (b) (i) Find values of A , B and C so that

$$\frac{2x^2 + x + 9}{(x^2 + 4)(x + 1)} = \frac{Ax + B}{x^2 + 4} + \frac{C}{x + 1}$$

(ii) Hence find $\int_0^2 \frac{2x^2 + x + 9}{(x^2 + 4)(x + 1)} dx$ giving your answer in exact form

1 (c) (i) On an Argand Diagram, draw and shade the region R given by

$$|z - 2 - 2i| \leq 2$$

2 (ii) P is a point in R , representing the complex number z . What is the maximum value of $|z|$?

2 (iii) The tangent to the curve at P cuts the x -axis at the point T .

By using the nature of $\triangle OPT$, or otherwise, find the exact area of $\triangle OPT$.

(d) Let $x = \alpha$ be a root of the polynomial $P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$
Where $(B + 2)^2 \neq 4A^2$

1 (i) Show that α cannot be 0, 1 or -1

1 (ii) Show that $\frac{1}{\alpha}$ is a root of $P(x) = 0$

2 (iii) Deduce that if α is a multiple root of $P(x) = 0$, then its multiplicity is 2

QUESTION 12: (Start a new page)

Marks

2 (a) Find the value of $\int_0^1 \tan^{-1} x \, dx$

(b) Let $f(x) = \ln(1+x) - \ln(1-x)$ where $-1 < x < 1$

1 (i) Show that $f'(x) > 0$ for all x in the given Domain

3 (ii) On the same diagram, sketch

$$y = \{\ln(1+x) \text{ for } x > -1$$

$$y = \{\ln(1-x) \text{ for } x < 1$$

$$y = \{f(x) \text{ for } -1 < x < 1$$

clearly labelling all 3 graphs

1 (iii) Find an expression for the inverse function $y = f^{-1}(x)$

1 (c) (i) Show that $(1+t^2)^{n-1} + t^2(1+t^2)^{n-1} = (1+t^2)^n$

3 (ii) $I_n = \int_0^x (1+t^2)^n dt$ for $n = 1, 2, 3, \dots$

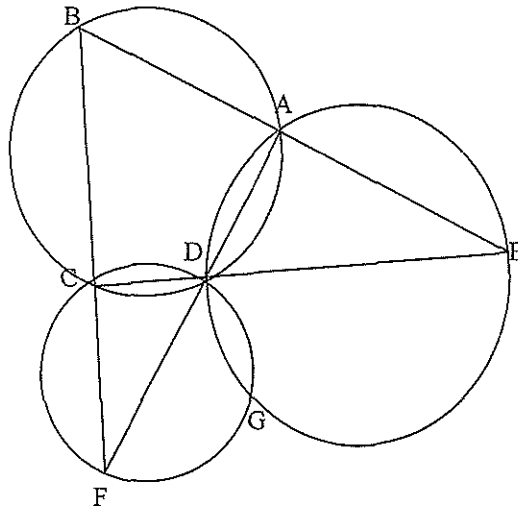
Use integration by parts, and part (i) above, to show that

$$I_n = \frac{1}{2n+1} (1+x^2)^n x + \frac{2n}{2n+1} I_{n-1}$$

Question 12 continues overpage.....)

QUESTION 12 continued.....)

4 (d)



In the diagram, ABCD is a cyclic quadrilateral.
BA and CD are produced to meet at E
Similarly BC and AD are produced to meet at F
Circles are then drawn through A, D and E, and C, D and F
These two circles intersect at D and G as shown

A copy of this diagram is included after your table of standard integrals.
Detach it, put it with your answer sheets, and then join

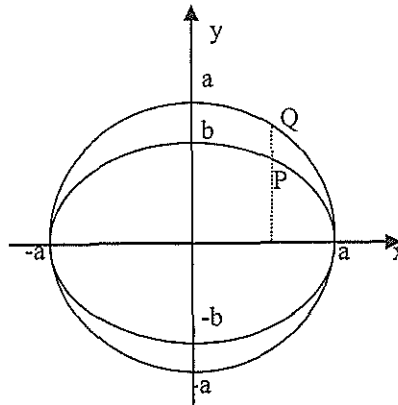
F to G
G to E and
D to G

Prove that E, G and F are collinear, clearly stating all geometric reasoning.

QUESTION 13: (Start a new page)

Marks

- (a) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$ where $a > b$, are drawn below.



P is a point on the ellipse with co-ordinates $(a\cos\theta, b\sin\theta)$.

A line perpendicular to the major axis is drawn through P to meet the circle at the point Q

- 1 (i) Find the co-ordinates of the point Q
- 2 (ii) Show that the equation of the tangent to the ellipse at P is given by

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

- 1 (iii) Find the equation of the tangent to the circle at Q
- 2 (iv) The tangents at P and Q meet at the point T.

Show that the point T lies on the x-axis.

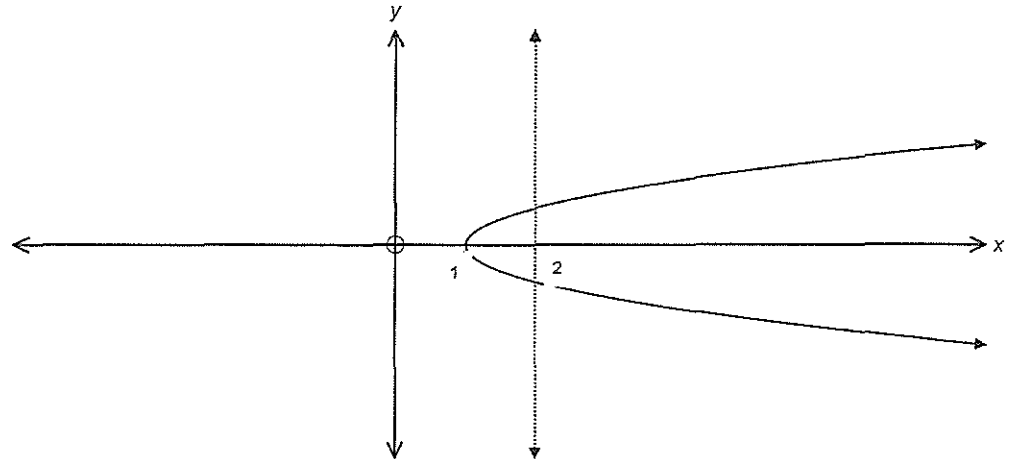
- 3 (b) $x^3 + px^2 + qx + r = 0$ has roots of α , β , and γ , where $\alpha = \beta + \gamma$
- Show that $p^3 - 4pq + 8r = 0$

Question 13 continues overpage.....)

QUESTION 13 continued.....)

2 (c) (i) Show that $\int x\sqrt{x-1}dx = \frac{2}{5}(x-1)^{\frac{5}{2}} + \frac{2}{3}(x-1)^{\frac{3}{2}} + c$

- 4 (ii) The area between the curve $y^2 = x - 1$ and the line $x = 2$, is rotated through 2π radians about the y axis



Using the method of cylindrical shells, taken parallel to the y -axis, show that the volume of the solid so formed is $\frac{64\pi}{15}$ cubic units.

QUESTION 14: (Start a new page)

Marks

- (a) The roots of the equation $z^5 + 1 = 0$ are $-1, \omega_1, \omega_2, \omega_3, \omega_4$ in cyclic order, anticlockwise around the Argand Diagram.

2 (i) Show that $\omega_1 = \overline{\omega_4}$

- 2 (ii) Find values of a, b and c so that $(z + 1)(z^4 + az^3 + bz^2 + cz + 1) = z^5 + 1$
and hence show that if ω is a root of $z^5 + 1 = 0$, not equal to -1 , then

$$\omega^4 + \omega^2 + 1 = \omega^3 + \omega$$

1 (iii) Show that $\omega_1^3 = \omega_3$

(For the rest of this question you may also assume the other results:

$$\omega_2^3 = \omega_1, \quad \omega_4^3 = \omega_2 \quad \text{and} \quad \omega_3^3 = \omega_4)$$

1 (iv) Deduce that $\omega_1^3 + \omega_2^3 + \omega_3^3 + \omega_4^3 = 1$

- 3 (v) By using the sum of the roots of $z^5 + 1 = 0$ in pairs, or otherwise, prove that

$$\cos \frac{4\pi}{5} + \cos \frac{2\pi}{5} = -\frac{1}{2}$$

- (b) $T_1(t, \frac{1}{t})$ and $T_2(3t, \frac{1}{3t})$ are two points on the hyperbola $xy=1$

2 (i) Show that, as t varies, the the midpoint of T_1T_2 lies on $3xy = 4$

1 (ii) Show that equation of the normal to the hyperbola $xy = 1$ at T_1 is given by

$$t^4 - t^3x + ty - 1 = 0$$

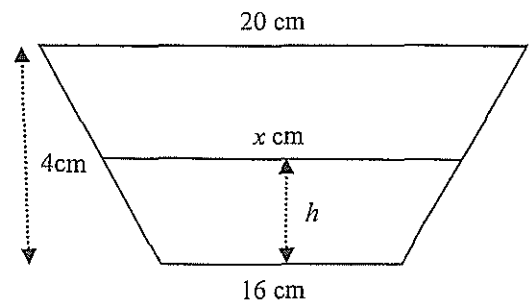
- 3 (iii) $R(0, h)$ is a point on the y -axis ($h \neq 0$). Show that there are exactly two points on $xy = 1$ with normals which pass through R .

QUESTION 15: (Start a new page)

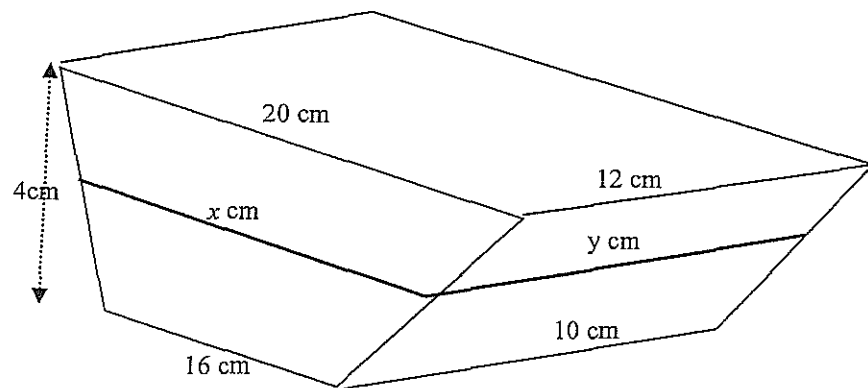
Marks

- (a) An isosceles trapezium has parallel sides of 20cm and 16cm and a height of 4cm.

A line, parallel to the base, is taken h cm above the 16 cm side, and has length x cm.



- 2 (i) By considering the *areas* of the three trapezia thus formed, or otherwise, prove that $x = 16 + h$
- 4 (ii) A cake tin is made using the shape above as its two ends, and two more equal trapezia as shown for its two sides.



The strip corresponding to x cm along the sides is of length y cm and you may assume the result $y = 10 + \frac{h}{2}$

Find the volume of the cake tin. (Show all appropriate working.)

Question 15 continues overpage.....)

QUESTION 15 continued.....)

- (b) A particle of mass 1 Kg is projected vertically upwards from the ground with a speed of 20m/s. The particle is under the effect of both gravity(g) and an air resistance of magnitude $\frac{1}{40}v^2$ where v is the current velocity of the particle at any time. The effects of gravity and air resistance remain the same no matter which direction the particle is going.

- 1 (i) Explain why the acceleration of the particle at any time whilst travelling upwards is given by:

$$\ddot{x} = -g - \frac{1}{40}v^2$$

(For the remainder of this question you may use $g = 10 \text{ m/s}^2$)

- 4 (ii) Calculate the greatest height reached by the particle

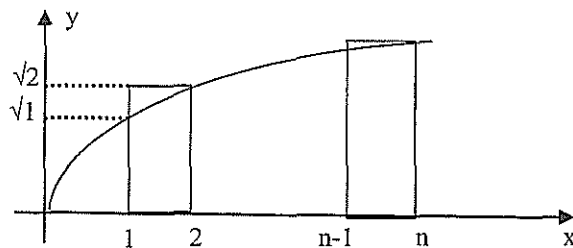
- 1 (iii) Write an expression for the acceleration of the particle as it returns to earth.

- 3 (iv) Find the speed of the particle *just before* it strikes the ground.

QUESTION 16: (Start a new page)

Marks

- (a) The figure below is of the curve $y = \sqrt{x}$. It is not drawn to scale.



- 1 (i) Show that the curve is increasing for all $x \geq 0$
- 2 (ii) Referring to the diagram above, show that $\sqrt{1} + \sqrt{2} + \dots + \sqrt{n} > \frac{2}{3}n\sqrt{n}$
for all finite values of $n \geq 1$

- 1 (iii) Prove, by expansion, or otherwise, that:

$$(4n + 3)^2 n < (4n + 1)^2 (n + 1)$$

- 4 (iv) Use Mathematical Induction to show that

$$\sqrt{1} + \sqrt{2} + \dots + \sqrt{n} < \frac{4n+3}{6} \sqrt{n} \quad \text{for all integers } n \geq 1$$

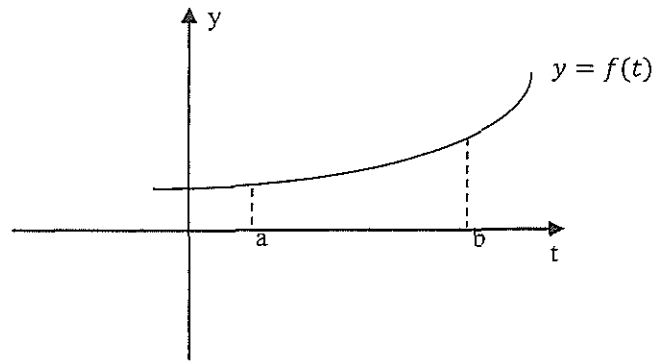
- 1 (v) Using parts (ii) and (iv) estimate

$$\sqrt{1} + \sqrt{2} + \dots + \sqrt{10\,000} \quad \text{to the nearest hundred.}$$

Question 16 continues overpage.....)

QUESTION 16 continued.....)

- 2 (b) (i) Let m and M be the smallest and greatest values of the integrable function $f(t)$ in the Domain $a \leq t \leq b$, as shown in the diagram below:



Explain carefully why

$$m(b - a) \leq \int_a^b f(t) dt \leq M(b - a)$$

- 3 (ii) Using part (i), or otherwise, deduce that,

$$\text{if } x > 0, \quad \frac{x}{1+x} \leq \log(1+x) \leq x$$

- 1 (iii) Hence show that $1 \leq \ln 4 \leq 2$

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

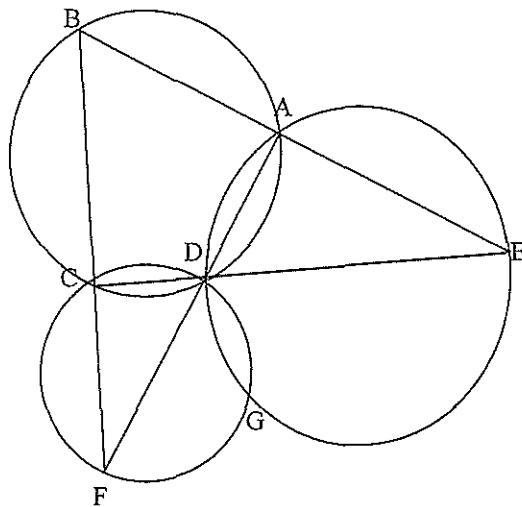
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

THIS DIAGRAM IS FOR QUESTION 12 (d)

It should be removed from the question booklet and placed with your answer booklet.

12 (d)



In the diagram, ABCD is a cyclic quadrilateral.
BA and CD are produced to meet at E
Similarly BC and AD are produced to meet at F
Circles are then drawn through A, D and E, and C, D and F
These two circles intersect at D and G as shown

Join:

F to G
G to E and
D to G

STHS EXTENSION 2 TRIAL HSC

AUGUST 2012

QUESTION

- | | |
|----|---|
| 1 | D |
| 2 | B |
| 3 | D |
| 4 | C |
| 5 | A |
| 6 | D |
| 7 | B |
| 8 | D |
| 9 | A |
| 10 | C |

MARKING-

QUESTION 11:

$$(a) \int \frac{dx}{x^2 - 6x + 13} = \int \frac{dx}{(x-3)^2 + 4}$$

$$= \frac{1}{2} \tan^{-1} \frac{x-3}{2} + k$$

2 MARKS

1 for inverse tan

1 for $\frac{1}{2}$. No penalty for

$$(b) (i) (Ax+B)(x+1) + C(x^2+4) = 2x^2 + x + 9$$

$$x = -1 \quad 5C = 10$$

$$C = 2$$

Coefficients of x^2 $A + C = 2$

$$\therefore A = 0$$

Constants $B + 4C = 9$

$$B = 1$$

$$\begin{cases} A = 0 \\ B = 1 \\ C = 2 \end{cases}$$

2 MARKS

1 off for each of
A, B, C incorrect

$$(ii) \int_0^2 \frac{2x^2 + x + 9}{(x^2 + 4)(x+1)} dx = \int_0^2 \frac{1}{x^2 + 4} dx + \int_0^2 \frac{2}{x+1} dx$$

$$= \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2 + 2 \ln(x+1) \Big|_0^2$$

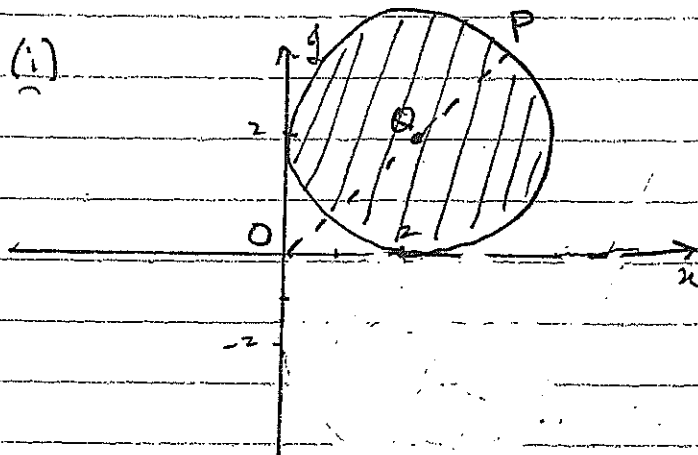
$$= \frac{1}{2} \tan^{-1} \frac{1}{2} + 2 \ln 3$$

$$= 2 \ln 3 + \frac{\pi}{8}$$

2 MARKS

1 for each part
of answer.

Q 11 (c) (i)



MARKING

1 MARK

(ii)

$$OQ = \sqrt{8} \quad (\text{by Pythagoras' Theorem})$$

$$\therefore OP = 2 + 2\sqrt{2}$$

$$\therefore \max |z| = 2 + 2\sqrt{2}$$

(iii)

There is a right angle at P

$$\angle TOP = \angle OTP = 45^\circ \Rightarrow \text{isosceles } \triangle OPT$$

$$\therefore PT = OP$$

$$\begin{aligned} \therefore \text{Area } \triangle OPT &= \frac{1}{2} \cdot OP \cdot PT \\ &= \frac{1}{2} (2 + 2\sqrt{2})^2 \\ &= 2(3 + 2\sqrt{2}) \end{aligned}$$

1 MARK

1 MARK

1 MARK for
realising this

(OR OTHERWISE)

1 MARK

(d)

(i) $x = \alpha$ is a root of $P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$

$$P(0) \neq 0 \Rightarrow \alpha \text{ cannot be } 0$$

$$P(1) = 2 + 2A + B = 0 \text{ only if } (B + 2)^2 = 4A^2$$

$$P(-1) = 2 - 2A + B = 0$$

1 MARK

(ii)

$$P\left(\frac{1}{\alpha}\right) = \frac{1}{\alpha^4} + \frac{A}{\alpha^3} + \frac{B}{\alpha^2} + \frac{A}{\alpha} + 1$$

$$= \frac{1}{\alpha^4} (1 + A\alpha + B\alpha^2 + A\alpha^3 + \alpha^4)$$

$$= 0 \text{ since } P(\alpha) = 0$$

1 MARK

(iii)

If α has multiplicity N then α does $\frac{1}{\alpha}$
(because of (ii) above)

Because $P(x)$ is of degree 4 it has
at most 4 roots.

$\therefore$ MAX value of N is 2.

2 for reasoning

QUESTION 12

MARKING

$$\begin{aligned}
 (a) \quad \int_0^1 \tan^{-1} x \, dx &= \int_0^1 \frac{d}{dx}(x) \tan^{-1} x \, dx \\
 &= x \tan^{-1} x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx \\
 &= \tan^{-1}(1) - \frac{1}{2} \ln(1+x^2) \Big|_0^1 \\
 &= \frac{\pi}{4} - \frac{1}{2} \ln 2
 \end{aligned}$$

2 MARKS

$$(b) \quad f(x) = \ln(1+x) - \ln(1-x), \quad -1 < x < 1$$

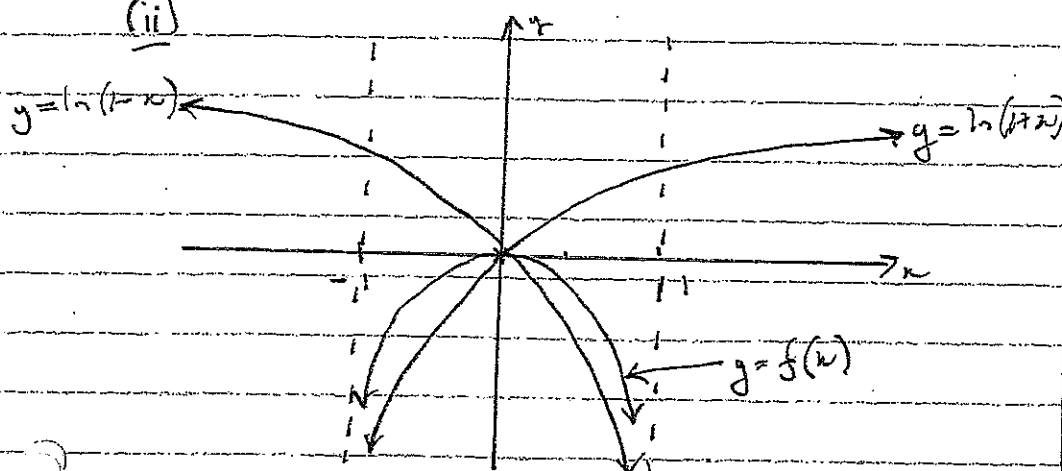
$$(i) \quad f'(x) = \frac{1}{1+x} + \frac{1}{1-x}$$

$$= \frac{1-x+1+x}{1-x^2}$$

$$= \frac{2}{1-x^2} > 0 \quad \forall -1 < x < 1$$

1 for this

(ii)



1 MARK for
each graph
= (3)

$$(iii) - \quad y = \ln\left(\frac{1+x}{1-x}\right)$$

Assumes $x = \ln\left(\frac{1+y}{1-y}\right)$

$$\frac{1+y}{1-y} = e^x$$

$$1+y = e^x - ye^x$$

$$y(1+e^x) = e^x - 1$$

$$y = \frac{e^x - 1}{e^x + 1}$$

$$\therefore f'(x) = \frac{e^x - 1}{e^x + 1}$$

1 MARK

MARKING

$$\begin{aligned} (c) \quad & (1+t^2)^{n-1} + t^2(1+t^2)^{n-1} \\ &= (1+t^2)^{n-1} [1+t^2] \\ &= (1+t^2)^n \end{aligned}$$

1 MARK

$$(ii) \quad \int_0^x (1+t^2)^n dt = t(1+t^2)^n \Big|_0^x - \int_0^x 2t^2(n)(1+t^2)^{n-1} dt$$

1 for this

$$\begin{aligned} \text{Now} \quad & \int_0^x 2t^2 n(1+t^2)^{n-1} dt \\ &= 2n \int_0^x t^2(1+t^2)^{n-1} dt \end{aligned}$$

$$= 2n \int_0^x (1+t^2)^n dt - 2n \int_0^x (1+t^2)^{n-1} dt$$

from part (i)

1 for "seeing" this.

$$\therefore I_n = t(1+t^2)^n \Big|_0^x - 2n I_n + 2n I_{n-1}$$

1 for completion

$$\therefore I_n(2n+1) = x(1+x^2)^n - 2n I_{n-1}$$

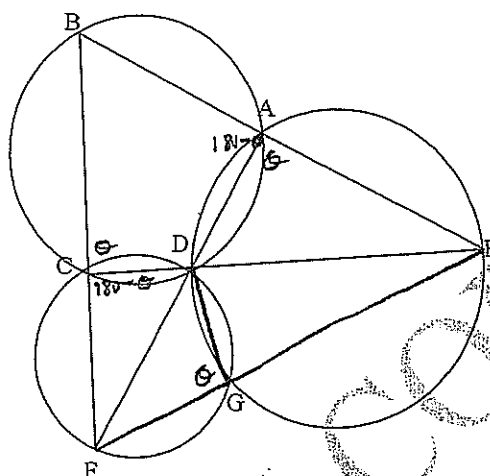
$$\therefore I_n = \frac{x(1+x^2)^n}{2n+1} + \frac{2n}{2n+1} I_{n-1}$$

(d) See attachment

THIS DIAGRAM IS FOR QUESTION 12 (d)

It should be removed from the question booklet and placed with your answer booklet.

12 (d)



In the diagram, ABCD is a cyclic quadrilateral.
BA and CD are produced to meet at E
Similarly BC and AD are produced to meet at F
Circles are then drawn through A, D and E, and C, D and F
These two circles intersect at D and G as shown

Join:

F to G
G to E and
D to G

$$\text{Let } \angle FGD = \theta^\circ$$

$$\therefore \angle FCD = (180 - \theta)^\circ \quad (\text{opposite angles of cyclic quadrilateral } FCDG)$$

$$\therefore \angle BCD = \theta^\circ \quad (\text{straight angle } BCF)$$

$$\therefore \angle BAD = (180 - \theta)^\circ \quad (\text{opposite angles of cyclic quadrilateral } CDAB)$$

$$\therefore \angle EAD = \theta^\circ \quad (\text{straight angle } BAE)$$

$$\therefore \angle DGE = (180 - \theta)^\circ \quad (\text{opposite angles of cyclic quadrilateral } AEGD)$$

$$\therefore \angle DGE + \angle FGD = 180^\circ$$

$\therefore$ FGE is a straight line.

① →

QUESTION 13:

(a) (i) Q is $(a \cos \theta, a \sin \theta)$

← 1 MARK

(ii) $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = -\frac{2y}{a^2} \cdot \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

At P $m_T = \frac{-b \cos \theta}{a \sin \theta}$

← 1 MARK

Equation of tangent:

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$ay \sin \theta + bx \cos \theta = ab \quad (1)$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

} 1 for correct working

(iii) At Q $\frac{dy}{dx} = -\frac{x}{y}$

$$m_T = \frac{-\cos \theta}{\sin \theta}$$

Equation of tangent:

$$y - a \sin \theta = \frac{-\cos \theta}{\sin \theta} (x - a \cos \theta)$$

$$y \sin \theta - a \sin^2 \theta = -x \cos \theta + a \cos^2 \theta$$

$$x \cos \theta + y \sin \theta = a \quad (2)$$

← 1 MARK

(iv) (1) - (2) $\frac{ay \sin \theta}{b} - y \sin \theta = 0$

$$\therefore \sin \theta \left(\frac{a}{b} - 1 \right) = 0$$

$$\text{Since } a \neq b \text{ and } \theta \neq 0^\circ$$

$$\therefore y = 0$$

← 1 MARK

} 1 MARK

(b)

$$x^3 + px^2 + qx + r = 0 \quad \alpha = \beta + \gamma$$

$$\text{Sum of roots} = 2\alpha = -p$$

$$\therefore \alpha = -\frac{p}{2}$$

← 1 for α

$$\text{Sum of roots} \times 2 \quad \alpha\beta + \alpha\gamma + \beta\gamma = q$$

$$\therefore -\frac{p}{2}(\beta + \gamma) + \beta\gamma = q$$

$$\frac{p^2}{4} + \beta\gamma = q$$

← 1 for $\beta\gamma$

$$\text{Product of roots} \quad \alpha\beta\gamma = -r$$

$$\therefore -\frac{p}{2} \left(q - \frac{p^2}{4} \right) = -r$$

$$\frac{p^3}{8} - \frac{pq}{2} = -r$$

$$p^3 - 4pq + 8r = 0$$

1 for completion

Q 13 CONTINUED

MARKING

$$\begin{aligned}
 (c)(i) \quad \int x\sqrt{x-1} \, dx &= \int (x-1)\sqrt{x-1} \, dx + \int \sqrt{x-1} \, dx \\
 &= \int (x-1)^{3/2} \, dx + \int (x-1)^{1/2} \, dx \\
 &= \frac{2}{5}(x-1)^{5/2} + \frac{2}{3}(x-1)^{3/2} + C
 \end{aligned}$$

} ①

} ①

OR let $u = \sqrt{x-1} \Rightarrow x = u^2 + 1$
 $dx = 2u \, du$

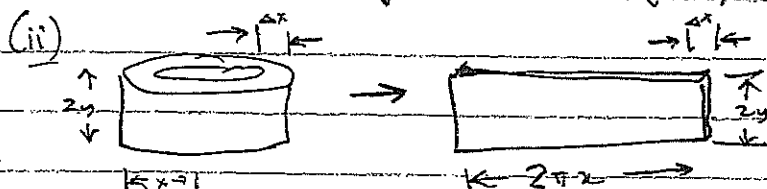
OR

∴ Integral = $\int (u^2 + 1) u \cdot 2u \, du$

← ①

$$\begin{aligned}
 &= \int 2u^4 \, du + \int 2u^2 \, du \\
 &= \frac{2}{5}u^5 + \frac{2}{3}u^3 + C \\
 &= \frac{2}{5}(x-1)^{5/2} + \frac{2}{3}(x-1)^{3/2} + C
 \end{aligned}$$

← ①



$$\begin{aligned}
 AV &= 2\pi x \cdot 2y \cdot \Delta x \\
 &= 4\pi xy \Delta x
 \end{aligned}$$

∴ Vol = $4\pi \int_1^2 xy \, dx$

← 2 MARKS to here

Since $y = \sqrt{x-1}$

$$Vol = 4\pi \int_1^2 x\sqrt{x-1} \, dx$$

$$= 4\pi \left[\frac{2}{5}(x-1)^{5/2} + \frac{2}{3}(x-1)^{3/2} \right]_1^2 \quad \text{from (i)} \quad \text{① for seeing this}$$

$$= 4\pi \left(\frac{2}{5} + \frac{2}{3} \right)$$

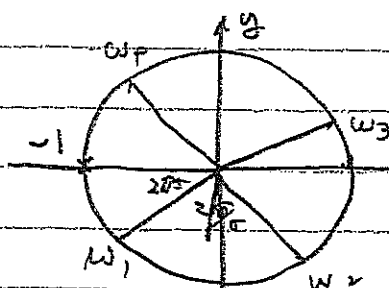
$$= \frac{64\pi}{15} \text{ or units}$$

1 MARK

QUESTION 14

MARKING

(a)



Angles between roots are $(2\pi/5)^\circ$

$$w_1 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$w_2 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

$$w_3 = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} \text{ or } \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5}$$

$$w_4 = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}$$

(i) $\bar{w}_4 = \cos \frac{8\pi}{5} - i \sin \frac{8\pi}{5}$

$$= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} = w_1$$

1 for these or similar.

PLVS

1 for this or similar.

= 2 MARKS

(ii) $a = -1$ $b = 1$ $c = -1$

← 1 MARK

Roots of $z^5 + 1 = 0$

are roots of $(z+1)(z^4 - z^3 + z^2 - z + 1) = 0$

Since $z \neq -1$ w solves $z^4 - z^3 + z^2 - z + 1 = 0$

$$w^4 - w^3 + w^2 - w + 1 = 0$$

$$\therefore w^4 + w^2 + 1 = w^3 + w$$

(iii) $w_1^3 = (\cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5})^3$

$$= \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$$

$$= \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5} \text{ because of unit circle}$$

$$= w_3$$

1 for this step

no MARKS here

1 MARK or

MUST carry some form of reasoning

QUEST 15 CONT...

$$\therefore x = -20 \log(40g + v^2) + 20 \log(40g + 400)$$

Since $g = 10$

$$x = -20 \log\left(\frac{800}{400 + v^2}\right)$$

At greatest height $v = 0$

$$\therefore x = 20 \log 2$$

1 MARK

1 MARK (or equivalent)

(iii) EARTH BOUND $\ddot{x} = g - \frac{1}{40}v^2$

1 MARK.

(iv) Restating the motion with $v=0, x=0$

$$v \frac{dv}{dx} = g - \frac{1}{40}v^2$$

$$\therefore \frac{dv}{dx} = \frac{40g - v^2}{40v}$$

$$\frac{dx}{dv} = \frac{40v}{400 - v^2} \quad \text{since } g = 10$$

$$\therefore x = -20 \log(400 - v^2) + C_2$$

At $v=0, x=0$

$$\therefore C_2 = 20 \log 400$$

$$\therefore x = 20 \log\left(\frac{400}{400 - v^2}\right)$$

← 1 MARK

← 1 MARK

At $x = 20 \log 2$ from pt (ii)

$$20 \log 2 = 20 \log\left(\frac{400}{400 - v^2}\right)$$

$$\therefore 2 = \frac{400}{400 - v^2}$$

$$\therefore 800 - 2v^2 = 400$$

$$\therefore v^2 = 200$$

$$\therefore v = 10\sqrt{2} \text{ m/s}$$

1 MARK

QUESTION 15:

$$(i) A_{\text{large}} = \frac{1}{2} \times 4(36)$$

$$A_1 = \frac{1}{2}h(x+16) \quad A_2 = \frac{1}{2}(20+x)(4-h)$$

$$h(x+16) + (20+x)(4-h) = 144$$

$$\therefore hx + 16h + 80 - 20h + 4x - xh = 144$$

$$-4h + 80 + 4x = 144$$

$$4x = 64 + 4h$$

$$x = 16 + h$$

(ii)

Volume of the "strip"

$$= xy \Delta h$$

$$\therefore \text{VOL}_{\text{tin}} = \lim_{\Delta h \rightarrow 0} \sum_{i=0}^4 xy \Delta h$$

$$= \int_0^4 xy \, dh$$

$$\text{Now } x = 16+h \text{ and } y = 10 + \frac{h}{2}$$

$$\therefore \text{VOL} = \int_0^4 160 + 8h + 10h + \frac{h^2}{2} \, dh$$

$$= \left[160h + 4h^2 + \frac{h^3}{6} \right]_0^4$$

$$= 640 + 144 + \frac{64}{6}$$

$$= 794 \frac{2}{3} \text{ cm}^3$$

$$(b)(i) \underline{t=0, x=0, m=1, v=20}$$

Two forces are acting against its upwards motion gravity (mg) and air resistance ($\frac{1}{40}v^2$). Since $m=1$

$$\ddot{x} = -g - \frac{1}{40}v^2$$

$$(ii) v \frac{dv}{dx} = -g - \frac{v^2}{40}$$

$$\frac{dv}{dx} = \frac{-40g - v^2}{40v}$$

$$\therefore \frac{dx}{dv} = \frac{-40v}{40g + v^2}$$

$$x = -20 \log(40g + v^2) + C_1$$

$$\text{At } x=0, v=20$$

$$\therefore C_1 = +20 \log(40g + 400)$$

①

①

① using "h"

①

② for limits

① should mention it is against the direction of motion

①

①

Q 13 (iv) In $z^5 + 1 = 0$

Sum of roots = 0

$\therefore w_1 + w_2 + w_3 + w_4 + -1 = 0$

$w_2^3 + w_4^3 + w_1^3 + w_3^3 = 1$

(v) Sum of roots in pairs = 0

i.e.

$-w_1 - w_2 - w_3 - w_4$

$+ w_1 w_2 + w_1 w_3 + w_1 w_4$

$+ w_2 w_3 + w_2 w_4$

$+ w_3 w_4 = 0$

i.e. $-1 + \cos \frac{7\pi}{5} \cos \frac{9\pi}{5} + \cos \frac{7\pi}{5} \cos \frac{\pi}{5} + \cos \frac{7\pi}{5} \cos \frac{3\pi}{5}$
 $+ \cos \frac{9\pi}{5} \cos \frac{11\pi}{5} + \cos \frac{9\pi}{5} \cos \frac{3\pi}{5}$
 $+ \cos \frac{11\pi}{5} \cos \frac{3\pi}{5} = 0$

$\therefore -1 + \cos \frac{6\pi}{5} + \cos \frac{3\pi}{5} + \cos 2\pi$
 $+ \cos 2\pi + \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = 0$

$\therefore 1 + \cos \left(-\frac{4\pi}{5} \right) + \cos \frac{3\pi}{5} + \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = 0$

$\therefore 2 \cos \frac{4\pi}{5} + 2 \sin \frac{2\pi}{5} = -1$

$\therefore \cos \frac{4\pi}{5} + \sin \frac{2\pi}{5} = -\frac{1}{2}$

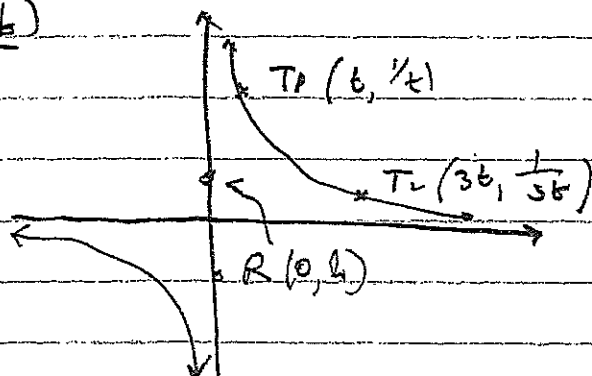
3 marks

① for getting angles to be between 0 and π

② for getting all to be in terms of $\frac{2\pi}{5}$ and $\frac{4\pi}{5}$

③ for "converting" cos and sin's

(b)



(i) midpoint is $M = \left(\frac{t+3t}{2}, \frac{\frac{1}{t} + \frac{1}{3t}}{2} \right)$

$\therefore M = \left(2t, \frac{4}{6t} \right)$

$x = 2t \quad y = \frac{2}{3t}$

$\therefore y = \frac{2}{3} \left(\frac{1}{2} \right)$

$\frac{xy}{2} = \frac{2}{3}$

$3xy = 4$

(ii)

$\frac{dy}{dx} = -\frac{1}{x^2}$

At T_1

$m_T = -\frac{1}{t^2}$

Equation of normal $y - \frac{1}{t} = t^2(x - t)$

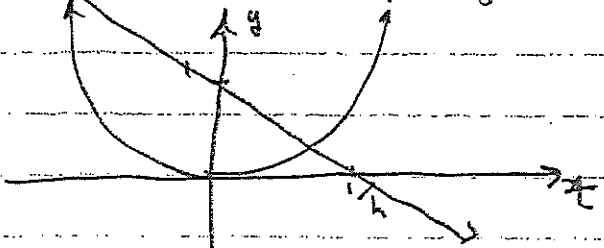
$t, y - 1 = t^3x - t^4$

i.e. $t^4 - t^3x + t, y - 1 = 0$

(iii) If R is on the normal, R satisfies the equation above.

$\therefore t^4 + t, h - 1 = 0$

Graphing $y = t^4$ and $y = 1 - t, h$ to look at pts of intersection



This only ever has 2 solutions

($\frac{1}{h}$ cannot be on y-axis as $\frac{1}{h} \neq 0$)

← ①

} for eliminating t

} ①

①

} ② or similar

QUESTION 1b

(a) (i) $\frac{dy}{dx} = \frac{1}{2}x^{-1/2}$
 $= \frac{1}{2\sqrt{x}} > 0 \quad \forall x > 0$

OR

At $x=n$, $y=\sqrt{n}$
 At $x=n+1$, $y=\sqrt{n+1} > \sqrt{n}$
 $\therefore$ increasing

either method
for 1 MARK

(ii) The rectangles drawn are all 1 unit wide

Area of the "big" rectangles
 $= 1 \times [\sqrt{1} + \sqrt{2} + \dots + \sqrt{n}]$

The "exact" area is given by

$\int_0^n \sqrt{x} dx = \left[\frac{2}{3}x^{3/2} \right]_0^n$
 $= \frac{2}{3}n^{3/2}$

1 for realising this
concept

And is less than the rectangles

$\therefore \sqrt{1} + \sqrt{2} + \dots + \sqrt{n} > \frac{2}{3}n^{3/2}$

1 MARK

(iii) $16n^3 + 24n^2 + 9n > 16n^3 + 24n^2 + 9n + 1$
 $= (16n^2 + 8n + 1)(n+1)$
 $= (4n+1)^2(n+1)$

1 for any correct
method

(iv) For $n=1$

LHS = $\sqrt{1}$ RHS = $7/6 >$ LHS

$\therefore$ the formula is true for $n=1$

← ① for testing
 $n=1$

Assume the formula is true for $n=k$

$\therefore \sqrt{1} + \sqrt{2} + \dots + \sqrt{k} < \frac{4k+3}{6} \sqrt{k}$

For $n=k+1$

$\sqrt{1} + \sqrt{2} + \dots + \sqrt{k} + \sqrt{k+1}$
 $< \frac{4k+3}{6} \sqrt{k} + \sqrt{k+1}$

(the result from part (iii) can be rewritten as:

$(4n+3)\sqrt{n} < (4n+1)\sqrt{n+1}$

① for realising the
connection

$\therefore \sqrt{1} + \sqrt{2} + \dots + \sqrt{k} + \sqrt{k+1} < \frac{1}{6}(4k+1)\sqrt{k+1} + \sqrt{k+1}$
 $= \frac{1}{6}\sqrt{k+1}(4k+7)$

← ② to get here

which is of the same form as for $n=k$

$\therefore$ If the formula is true for $n=k$ it is true for $n=k+1$

But it is true for $n=1$

" " " " $n=2$ and so on

ie true $\forall n$

QUESTION 16 CONT...

$$(v) \sqrt{1} + \sqrt{2} + \dots + \sqrt{10000} < \frac{40003}{6} \sqrt{10000}$$

from part (iv)

and from part (ii)

$$\sqrt{1} + \sqrt{2} + \dots + \sqrt{10,000} > \frac{2}{3} 10000 \sqrt{10,000}$$

$$\frac{4000300}{6} > \text{EXP} > \frac{2000000}{3}$$

$$\therefore \text{EXP} \approx 666,700$$

1 MARK

(b)(i) Area of small rectangle = $(b-a)m$.

which is less than the exact area = $\int_a^b f(t) dt$

which is less than the area

of the large rectangle = $(b-a)M$.

(ii) Let $y = \frac{1}{1+t}$ be the function above

while $a=0$ and $b=x$

1 MARK

$\therefore$ The smallest value of y is $\frac{1}{1+x}$
 " largest " " " is 1

$$2. (x-0) \frac{1}{1+x} \leq \int_0^x \frac{1}{1+t} dt \leq (x-0) 1$$

1 for here

$$\frac{x}{1+x} \leq \ln(1+x) \leq x$$

$$\frac{x}{1+x} \leq \ln(1+x) \leq x$$

1 for this

(iii) Set $x=1$

$\therefore$ from above $\frac{1}{2} \leq \ln 2 \leq 1$

Doubling all terms

$$1 \leq 2 \ln 2 \leq 2$$

$$\text{ie } 1 \leq \ln 4 \leq 2$$

1 MARK